Scalability & Future Plan

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being (HDI Predictor)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to a static historical HDI dataset	Cannot reflect very recent country-level changes in real time	Medium
2	Single-model approach without ensembling	May reduce prediction robustness on edge-case inputs	Medium
3	No user authentication or data storage	Limits personalization and historical tracking of predictions	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single-user local Flask requests	Deploy on cloud (AWS/Azure/Render) with load balancing for concurrent users
2	Data Storage	Static local CSV dataset	Migrate to a cloud database (MySQL/MongoDB) for dynamic, updatable data
3	Performance	Model loaded fresh per request	Cache model in memory and serve via Gunicorn/WSGI for faster inference
4	Security	No authentication or input validation layer	Add user login, HTTPS, and stricter input validation

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Add ensemble models (Random Forest + XGBoost)	1 Month	Improved prediction accuracy and robustness
Phase 3	Integrate real-time World Bank / UN data feed	2 Months	Live, continuously updated HDI predictions
Phase 4	Cloud deployment with authentication & analytics dashboard	3 Months	Scalable multi-user platform with usage insights